FRAUDTRACE AI

Account Investigation Case Report

Case field	Value
Case account	ACC002
Dataset	FraudTrace AI Synthetic Demo (demo_synthetic)
Generated at	2026-10-16 13:47 UTC
Graph traversal depth	3
Investigation label	medium
Transaction count	1
Flagged transaction count	1
Graph nodes	12
Graph relationships	15

Scope and limitation: This report summarizes the evidence returned by the FraudTrace AI demonstration system. A risk label or flagged transaction is a review signal, not a determination of criminal activity. Shared devices, IP addresses, or merchants can have legitimate explanations.

Investigation Finding

Multi-hop connected-account pattern found

System warning: Connections are investigation leads, not proof of fraud. Shared devices and IP addresses can have legitimate explanations.

Related Accounts

1. ACC001

2. ACC003

3. ACC006

Shared Device Signals

1. DEV001

2. DEV002

Multi-Hop Investigation Paths

start_account	shared_device	intermediary_account	shared_ip	connected_account	steps	evidence
ACC002	DEV002	ACC006	IP002	ACC003	["ACC002", "DEV002", "ACC006", "IP002", "ACC003"]	["ACC002 and ACC006 share device DEV002", "ACC006 and ACC003 share IP address IP002"]
ACC002	DEV002	ACC003	IP002	ACC006	["ACC002", "DEV002", "ACC003", "IP002", "ACC006"]	["ACC002 and ACC003 share device DEV002", "ACC003 and ACC006 share IP address IP002"]

Transaction Evidence

amount	merchant_id	status	transaction_id
8700	MER001	flagged	TXN002

Graph Investigation

Traversal depth: 3 | Returned nodes: 12 | Returned relationships: 15

Graph nodes

Node ID	Labels	Properties

ACC002	Account	{
		"id": "ACC002",
		"risk": "medium"
		}

TXN002	Transaction	{
		"id": "TXN002",
		"amount": 8700,
		"status": "flagged"
		}

MER001	Merchant	{
		"id": "MER001"
		}

TXN001	Transaction	{
		"id": "TXN001",
		"amount": 9500,
		"status": "flagged"
		}

IP002	IPAddress	{
		"id": "IP002"
		}

Node ID	Labels	Properties

ACC006	Account	{
		"id": "ACC006",
		"risk": "low"
		}

DEV002	Device	{
		"id": "DEV002"
		}

ACC003	Account	{
		"id": "ACC003",
		"risk": "high"
		}

TXN003	Transaction	{
		"id": "TXN003",
		"amount": 1200,
		"status": "normal"
		}

IP001	IPAddress	{
		"id": "IP001"
		}

ACC001	Account	{
		"id": "ACC001",
		"risk": "high"
		}

DEV001	Device	{
		"id": "DEV001"
		}

Graph relationships

Source	Relationship	Target
ACC002	MADE	TXN002
TXN002	PAID_TO	MER001
TXN001	PAID_TO	MER001
ACC002	CONNECTS_FROM	IP002
ACC006	CONNECTS_FROM	IP002
ACC006	USES_DEVICE	DEV002
ACC003	CONNECTS_FROM	IP002
ACC003	MADE	TXN003
ACC003	USES_DEVICE	DEV002
ACC002	CONNECTS_FROM	IP001
ACC001	CONNECTS_FROM	IP001
ACC001	MADE	TXN001
ACC001	USES_DEVICE	DEV001
ACC002	USES_DEVICE	DEV002
ACC002	USES_DEVICE	DEV001

Investigation Assistant

Question: Why is this account suspicious?

Assistant Findings & Analysis:

ACC002 is flagged as suspicious because the current graph evidence shows several risk indicators:

1. **Medium stored risk label** - The account's stored risk label is "medium," which already places it on a watch-list.
2. **Flagged high-value transaction** - ACC002 made a single **$8,700** transaction at merchant **MER001** that is marked **flagged** [EV-TXN-TXN002].
3. **Connections to other accounts through shared infrastructure** -

- **ACC001** shares a device (**DEV001**), an IP address (**IP001**), and the same merchant (**MER001**) with ACC002; this link has a high review priority [EV-LINK-ACC001].
- **ACC003** shares a device (**DEV002**) and an IP address (**IP002**) with ACC002 [EV-LINK-ACC003].
- **ACC006** also shares the same device (**DEV002**) and IP address (**IP002**) with ACC002 [EV-LINK-ACC006].

These shared resources (devices, IPs, merchant) and the flagged transaction are typical investigation leads in fraud analysis. While they do not prove wrongdoing, they collectively explain why ACC002 is considered suspicious in the current synthetic dataset.

Assistant Evidence

type	account_id	signals	shared_devices	shared_ips	common_merchants	sent_transactions
connected_account	ACC001	["shared_device", "shared_ip", "shared_merchant"]	["DEV001"]	["IP001"]	["MER001"]	[]
connected_account	ACC003	["shared_device", "shared_ip"]	["DEV002"]	["IP002"]	[]	[]
connected_account	ACC006	["shared_device", "shared_ip"]	["DEV002"]	["IP002"]	[]	[]
transaction	Not provided	Not provided	Not provided	Not provided	Not provided	Not provided

Assistant limitation: Synthetic demonstration data. Shared devices, IP addresses, merchants, and flagged transactions are investigation leads, not proof of fraud.

Audit Notes

This report was generated from the API responses available at the time of export. It is not a forensic certification or an automated determination of fraud. Analysts should verify source records, consider alternative explanations, and follow their organization's review and escalation procedures.